

Lecture 2

Span, Linear Independence, Basis, Dimension

Throughout, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Tags indicate the flavour of each problem; a $\star$ marks a harder one. Problems that ask whether a vector lies in a span or whether vectors are linearly independent, and problems that ask for the coordinates of a vector, all reduce to solving linear systems.

Core tutorial route

Work **2.1, 2.12, 2.13, 2.15, 2.17, 2.22, 2.26, and 2.27** in that order. This route covers span, field-sensitive independence, basis construction and extension, the linear-dependence lemma, coordinate calculation and proof, and a dimension-formula transfer.

Additional practice: 2.2–2.11, 2.14, 2.16, 2.18–2.21, 2.23–2.25, 2.28–2.30. **Bridge:** none required. **Extension:** 2.31.

Span

Exercise 2.1 (computational) Is $(1, 2, 3)$ in $\text{span}((1, 1, 0), (0, 1, 1))$ in $\mathbb{R}^3$?

Answer. No: the system is inconsistent, so $(1, 2, 3) \notin \text{span}((1, 1, 0), (0, 1, 1))$.

Exercise 2.2 (computational) Is $(2, 3, 1)$ in $\text{span}((1, 1, 0), (0, 1, 1))$ in $\mathbb{R}^3$? If so, write it as a linear combination.

Answer. Yes: $(2, 3, 1) = 2(1, 1, 0) + (0, 1, 1)$.

Exercise 2.3 (computational) For which value of t is $(t, 1, 2)$ in $\text{span}((1, 0, 1), (1, 1, 1))$ in $\mathbb{R}^3$?

Answer. $t = 2$.

Exercise 2.4 (computational) Does the polynomial $p(t) = 2 + 5t + 3t^2 \in \mathcal{P}_2(\mathbb{R})$ lie in the subspace $\text{span}(1 + t, t + t^2)$?

Answer. Yes: $2 + 5t + 3t^2 = 2(1 + t) + 3(t + t^2)$.

Exercise 2.5 (proof) Show that $\text{span}(v_1, \dots, v_m) = \text{span}(v_1, \dots, v_m, w)$ if and only if $w \in \text{span}(v_1, \dots, v_m)$.

Hint. One direction is immediate. For the other, substitute $w = \sum a_i v_i$ into a combination that uses w and collect the v_i .

Linear independence

Exercise 2.6 (computational) Find the value of t for which $(3, 1, 4), (2, -3, 5), (5, 9, t)$ are linearly dependent in $\mathbb{R}^3$.

Answer. $t = 2$.

Exercise 2.7 (computational) Are $(1, 3), (2, 5)$ linearly independent in $\mathbb{R}^2$?

Answer. Yes.

Exercise 2.8 (computational) Are $(1, 0, 1)$, $(2, 1, 0)$, $(3, 1, 2)$ linearly independent in $\mathbb{R}^3$?

Answer. Yes (the determinant is 1).

Exercise 2.9 (computational) Are $1 + t$, $1 - t$, t^2 linearly independent in $\mathcal{P}_2(\mathbb{R})$?

Answer. Yes.

Exercise 2.10 (computational) Are $(1, i)$, $(i, -1)$ linearly independent in $\mathbb{C}^2$?

Answer. No: they are linearly dependent.

Exercise 2.11 (proof) Suppose v_1, v_2, v_3, v_4 are linearly independent in V . Prove that

$$v_1 - v_2, \quad v_2 - v_3, \quad v_3 - v_4, \quad v_4$$

are also linearly independent.

Hint. Set a general combination equal to 0 and gather the coefficient of each v_i ; independence of the original list gives a system you can solve one variable at a time.

Exercise 2.12 (★) View $\mathbb{C}$ as a vector space. Show that $1 + i$, $1 - i$ are linearly independent over $\mathbb{R}$ but linearly dependent over $\mathbb{C}$.

Hint. The scalar field is doing the work. Over $\mathbb{R}$, separate real and imaginary parts; over $\mathbb{C}$, ask what multiplying $1 + i$ by i gives.

Bases and dimension

Exercise 2.13 (computational) Find a basis and the dimension of $U = \{x \in \mathbb{R}^4 : x_1 - x_2 + 2x_3 = 0, x_2 - x_3 + x_4 = 0\}$.

Answer. Basis $(-1, 1, 1, 0)$, $(-1, -1, 0, 1)$; $\dim U = 2$ (any basis of U is acceptable).

Exercise 2.14 (computational) Give a basis and the dimension of (a) the symmetric 2×2 real matrices, and (b) the trace-zero 2×2 real matrices.

Answer. (a) dimension 3; (b) dimension 3.

Exercise 2.15 (computational) Extend the independent list $(1, 2, 1, 0)$, $(0, 1, 1, 1)$ to a basis of $\mathbb{R}^4$.

Answer. E.g. $(1, 2, 1, 0)$, $(0, 1, 1, 1)$, $(1, 0, 0, 0)$, $(0, 0, 1, 0)$ (other choices work).

Exercise 2.16 (computational) Find a basis and the dimension of $U = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$.

Answer. Basis $(-1, 1, 0)$, $(-1, 0, 1)$; $\dim U = 2$.

Exercise 2.17 (computational) Find a basis and the dimension of $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 0\}$.

Answer. Basis t , t^2 , t^3 ; dimension 3.

Exercise 2.18 (computational) Find a basis and the dimension of the space U of diagonal 3×3 real matrices.

Answer. Basis $\text{diag}(1, 0, 0)$, $\text{diag}(0, 1, 0)$, $\text{diag}(0, 0, 1)$; $\dim U = 3$.

Exercise 2.19 (computational) Use the linear-dependence lemma to remove a redundant vector, then find the dimension of the subspace

$$U = \text{span}((1, 2, 3, 4), (2, 4, 6, 8), (1, 0, 1, 0), (0, 1, 0, 1)) \subseteq \mathbb{R}^4.$$

Answer. 3.

Exercise 2.20 (proof) Let V be finite-dimensional and let $U \subseteq V$ be a subspace with $\dim U = \dim V$. Prove that $U = V$.

Hint. Take a basis of U and count: it is an independent list of length $\dim V$ sitting inside V .

Exercise 2.21 (proof) Let u_1, u_2, u_3 be a basis of a vector space V . Prove that the vectors $u_1, u_1 + u_2, u_1 + u_2 + u_3$ also form a basis of V .

Hint. Set a combination of the three new vectors equal to 0 and collect the coefficients of u_1, u_2, u_3 ; the resulting system is triangular, so solve it from the bottom up. Then count the list.

Coordinates

Exercise 2.22 (computational) Find the coordinate vector $[v]_{\mathcal{B}}$ of $v = (1, 2, 3)$ in the basis $\mathcal{B} = ((1, 1, 0), (0, 1, 1), (0, 0, 1))$.

Answer. $[v]_{\mathcal{B}} = (1, 1, 2)$.

Exercise 2.23 (computational) Find the coordinate vector $[v]_{\mathcal{B}}$ of $v = (5, 1)$ in the basis $\mathcal{B} = ((1, 1), (1, -1))$.

Answer. $[v]_{\mathcal{B}} = (3, 2)$.

Exercise 2.24 (computational) Find the coordinate vector $[p]_{\mathcal{C}}$ of $p(t) = t^2 \in \mathcal{P}_2(\mathbb{R})$ in the basis $\mathcal{C} = (1, t - 1, (t - 1)^2)$.

Answer. $[p]_{\mathcal{C}} = (1, 2, 1)$.

Exercise 2.25 (computational) Find the coordinate vector $[v]_{\mathcal{B}}$ of $v = (2, 3, 4)$ in the basis $\mathcal{B} = ((1, 0, 0), (1, 1, 0), (1, 1, 1))$.

Answer. $[v]_{\mathcal{B}} = (-1, -1, 4)$.

Exercise 2.26 (proof) Let $\mathcal{B}$ be a basis of an n -dimensional space V . Prove that vectors $v_1, \dots, v_k \in V$ are linearly independent if and only if their coordinate vectors $[v_1]_{\mathcal{B}}, \dots, [v_k]_{\mathcal{B}}$ are linearly independent in $\mathbb{F}^n$.

Hint. The coordinate map is a bijection that respects sums and scalar multiples, so it carries a relation among the v_i to a relation among their coordinate vectors, and back.

Sums and dimension formula

Exercise 2.27 (computational) In $\mathbb{R}^4$ let

$$U = \text{span}((1, 1, 1, 1), (1, 0, 1, 0)) \quad \text{and} \quad W = \text{span}((0, 1, 0, 1), (1, 1, 0, 0)).$$

Find $\dim(U + W)$ and $\dim(U \cap W)$, and exhibit a basis of $U \cap W$.

Answer. $\dim(U + W) = 3$, $\dim(U \cap W) = 1$, $U \cap W = \text{span}((0, 1, 0, 1))$.

Exercise 2.28 (computational) In $\mathbb{R}^3$ let

$$U = \text{span}((1, 0, 0), (0, 1, 0)) \quad \text{and} \quad W = \text{span}((0, 1, 0), (0, 0, 1)).$$

Find $\dim(U + W)$ and $\dim(U \cap W)$, and give a basis of $U \cap W$.

Answer. $\dim(U + W) = 3$, $\dim(U \cap W) = 1$, $U \cap W = \text{span}((0, 1, 0))$.

Exercise 2.29 (computational) In $\mathbb{R}^4$ let

$$U = \text{span}((1, 0, 0, 0), (0, 1, 0, 0)) \quad \text{and} \quad W = \text{span}((0, 0, 1, 0), (0, 0, 0, 1)).$$

Find $\dim(U + W)$ and $\dim(U \cap W)$. Is the sum direct?

Answer. $\dim(U + W) = 4$, $\dim(U \cap W) = 0$; the sum is direct.

Exercise 2.30 (computational) In $\mathbb{R}^4$ let

$$U = \text{span}((1, 1, 0, 0), (0, 0, 1, 1)) \quad \text{and} \quad W = \text{span}((1, 0, 0, 1), (0, 1, 1, 0)).$$

Find $\dim(U + W)$ and $\dim(U \cap W)$, and exhibit a basis of $U \cap W$.

Answer. $\dim(U + W) = 3$, $\dim(U \cap W) = 1$, $U \cap W = \text{span}((1, 1, 1, 1))$.

Exercise 2.31 (★) Let V be finite-dimensional with $\dim V = n$, and let U, W be subspaces with $\dim U + \dim W > n$. Prove that $U \cap W \neq \{0\}$.

Hint. Apply the dimension formula to $U + W$, and use that $U + W$ is still a subspace of the n -dimensional V .